

ARIFUZZAMAN JOY

AI/ML Infrastructure Engineer

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SUMMARY

AI/ML infrastructure engineer shipping LLM and agent systems on serverless/decentralized GPUs. Core contributor to a 4.6k-star open-source AI platform (14 PRs merged). Published researcher (7 papers, 4 Q1 SCI, 75+ citations). Delivers failure-hardened GPU tooling, multi-agent orchestration, and evaluated RAG pipelines — cutting inference cost ~60% vs. managed APIs. Upwork Top Rated, 100% JSS. Python 7+ yrs, ML 5+.

EXPERIENCE

Core Contributor — Osmantic/ODS (DreamServer)

2025 — Present

Open-source self-hosted AI platform (4.6k stars). 14 PRs merged, 1 major PR under review.

- Authored a ~6,250-line P2P GPU deployment toolkit for Vast.ai (PR #983, under review) — hardened Bash installer covering GPU-tier detection, driver-mismatch repair, and 28 documented host-failure modes.
- Shipped health-check routing, gpu_backends validation, env-var hardening, node capabilities, and shell-injection fixes across 14 merged PRs — all ShellCheck-clean, hard-fail ACL, let-it-crash standard.

Contributor — casys-kaist/LLMServingSim

2026

- Merged PR #59 into KAIST CASys lab's LLM-serving simulator: RTX 4090 skew sweep and benchmark metadata extending GPU coverage for inference research.

AI/ML Engineer — Freelance (Upwork & Direct Clients)

2023 — Present

- Delivered production AI systems (LLM serving, diffusion pipelines, RAG chatbots) end-to-end. Upwork Top Rated, 100% Job Success Score.
- Migrated inference from always-on VMs to serverless GPUs (RunPod, Modal, Vast.ai) — cut cost ~60%, sub-200ms p95 latency on 7B–70B models via vLLM with GPTQ/AWQ quantization.
- Built multi-modal RAG with hybrid retrieval, citation tracking, and LLM-as-judge evaluation (Ragas). Built a GPU node broker MCP server for agent-driven compute provisioning.

Research Assistant — University of Rajshahi

Mar 2022 — May 2023

- Applied neural networks to optimize thermophotovoltaic efficiency (+18%), producing 4 Q1 journal publications (IF up to 7.1). First-author paper in Heliyon (Cell Press).

RELEVANT PROJECTS

LatentMAS-SLoRA — Multi-Agent Reasoning Framework2026

- Extended ICML 2026 Spotlight paper (arXiv:2511.20639) with dynamic LoRA adapters — officially featured as community extension #5 in Gen-Verse/LatentMAS (879 stars, HF #1 Paper of the Day).
- Agents exchange hidden states, not text: +12% accuracy, 2.7x faster inference, 63.6% token reduction, 75% VRAM savings via 4-role adapter swapping on single Qwen 2.5 base.

Multi-GPU Video Generation — Wan 2.22025

- Distributed inference for SOTA video generation (T2V/I2V/S2V) using FSDP on serverless clusters. Topology-aware placement with per-GPU VRAM capping. 3x throughput.

Serverless Diffusion Ecosystem2025

- End-to-end serverless workers for FLUX.2, Whisper, and custom diffusion models — Dockerized, CI/CD, autoscaled to \$0 idle. Includes LoRA training pipeline.

PUBLICATIONS

- **ML-assisted revelation of best-performing single heterojunction TPV cell.** *Sust. Energy Tech. & Assess.*, 2025. (Q1, IF 7.1)
- **ML-enabled performance exploration of AuCuSe4 in TPV cell.** *Solar Energy*, 2024. (Q1, IF 6.0)
- **Open-source transformers in medical imaging** — brain MRI, 99.6% acc. *IJACSA*, 2024.
- **AgInTe2 chalcopyrite thin-film solar cell** (first author). *Heliyon*, 2023. (Q1, IF 4.0)
- + 3 papers in PV simulation & speech processing. [Scholar](#) — 75+ citations, h-index 5.

EDUCATION

B.Sc. Electrical & Electronic Engineering — University of Rajshahi2017 — 2020

TECHNICAL SKILLS

Languages	Python (7+ yrs), Bash, SQL (5+ yrs), JavaScript
LLM & GenAI	GPT-4o, Claude 3.5/4, Llama 3, Qwen 2.5, Flux, SDXL, Wan 2.2
Inference & Serving	vLLM, TensorRT-LLM, ONNX, Triton, FastAPI, GPTQ/AWQ quantization
GPU & MLOps	CUDA, FSDP, RunPod, Modal, Vast.ai, Docker, Kubernetes, CI/CD
RAG & Agents	LangChain, LangGraph, Pinecone, ChromaDB, pgvector, MCP, Ragas, PEFT/LoRA
Core ML & Data	PyTorch, HuggingFace Transformers, Diffusers, PostgreSQL, Git, Prometheus